

MTL 732 - Financial Mathematics Project

Report

Comprehensive Option Pricing, Risk Analysis & Trading Strategies

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1 Model Comparison

1.1 Objective

Compute and compare the price of a European **call** option using three methods:

1. **Black–Scholes–Merton (BSM)** closed-form formula (benchmark).
2. **Cox–Ross–Rubinstein (CRR) Binomial Tree** with N steps.
3. **Monte Carlo** simulation under risk-neutral measure.

1.2 Assumptions (common to all methods unless stated otherwise)

- Frictionless markets, continuous trading (for BSM).
- Constant risk-free rate r and constant volatility σ .
- It follows geometric Brownian motion under the real-world model; under risk-neutral pricing we replace drift by r .
- Option is European (exercise only at maturity T).
- No dividends (extensions for known dividends are straightforward).

1.3 Formulas

BSM Under these assumptions the price of a European call is:

$$C_{BSM} = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

with

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

This is derived by risk-neutral expectation and log-normality of S_T . It is the model benchmark because it is closed-form and fast.

CRR Binomial Tree Discretize $[0, T]$ into N steps of size $\Delta t = T/N$. Define

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Construct terminal payoffs at nodes and perform backward induction:

$$V_{i,j} = e^{-r\Delta t} (pV_{i+1,j+1} + (1-p)V_{i+1,j}).$$

As $N \rightarrow \infty$ the CRR price converges to BSM under the CRR assumptions.

Monte Carlo (Risk-Neutral) Simulate M independent draws of standard normal Z_k and compute

$$S_T^{(k)} = S_0 \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z_k\right).$$

The Monte Carlo estimator for a call is:

$$\hat{C}_{MC} = e^{-rT} \cdot \frac{1}{M} \sum_{k=1}^M \max(S_T^{(k)} - K, 0).$$

Monte Carlo converges at rate $O(M^{-1/2})$; variance reduction (antithetic variates, control variates) helps substantially.

1.4 Comparison and Metrics

We compute absolute and relative differences w.r.t. BSM:

$$\Delta_{CRR} = C_{CRR} - C_{BSM}, \quad \Delta_{MC} = C_{MC} - C_{BSM}.$$

Relative errors:

$$\varepsilon_{CRR} = \frac{\Delta_{CRR}}{C_{BSM}}, \quad \varepsilon_{MC} = \frac{\Delta_{MC}}{C_{BSM}}.$$

These metrics quantify discretization error (CRR) and sampling error (MC). Typical patterns:

- Δ_{CRR} decreases as N increases (convergence to BSM).
- Δ_{MC} shrinks in expectation as M increases, with standard error $\approx \text{sd}(\text{discounted payoff})/\sqrt{M}$.

1.5 Plots / Figures

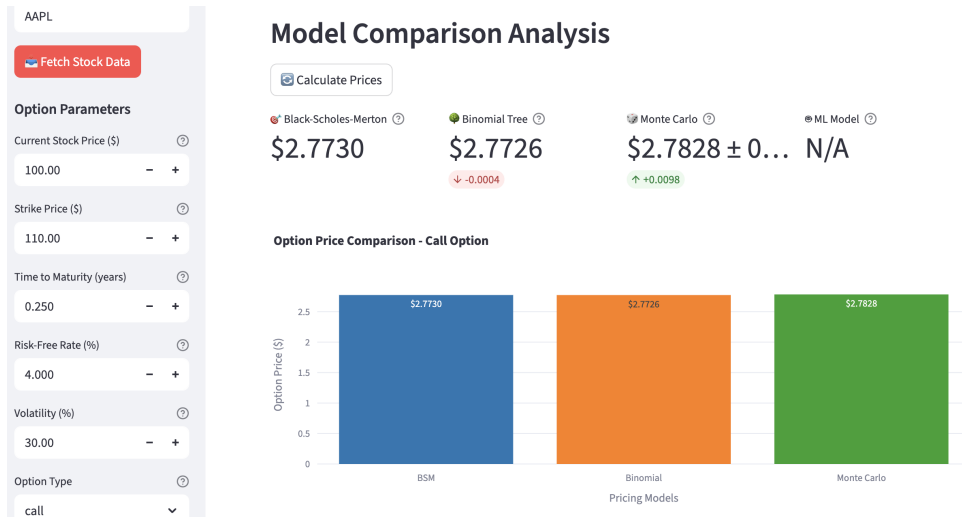


Figure 1: Example input set and comparison graph



Figure 2: Option price analysis for an example input set

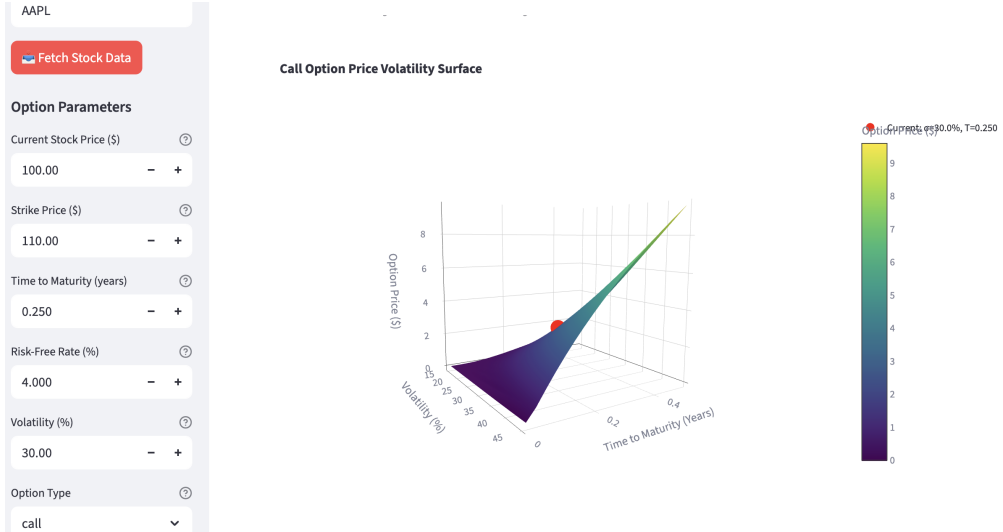


Figure 3: Call option implied volatility surface: volatility as a function of strike and time-to-maturity T .

2 Greek Analysis

2.1 Objective

This section analyzes the **risk sensitivities (Greeks)** of a European call option under the Black–Scholes–Merton (BSM) framework. Greeks measure how the option price responds to small changes in underlying factors such as stock price, volatility, time, and interest rate.

2.2 Formulas and Interpretation

Let d_1, d_2 be as defined in the BSM model:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

$\Delta = N(d_1)$ Sensitivity of option price to the underlying price S_0 .

$\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}}$ Rate of change of Δ with respect to S_0 .

$\Theta = -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}} - r K e^{-rT} N(d_2)$ Time decay — change in option value per unit time.

$\nu = S_0 N'(d_1) \sqrt{T}$ Sensitivity of option price to volatility σ .

$\rho = K T e^{-rT} N(d_2)$ Sensitivity of option price to risk-free rate r .

where $N(x)$ and $N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ denote the CDF and PDF of the standard normal distribution, respectively.

2.3 Intuitive Explanation

- **Delta** (Δ) — Measures how much the option price changes for a small change in the stock price. It represents the hedge ratio, i.e., how many shares are needed to hedge one option.
- **Gamma** (Γ) — Measures the curvature of the option price with respect to the stock price. A higher Gamma indicates the Delta changes rapidly, increasing hedging risk.
- **Theta** (Θ) — Measures the sensitivity of the option's value to time decay. A negative Theta indicates that the option loses value as time passes (holding other variables constant).
- **Vega** (ν) — Captures sensitivity to volatility. Higher Vega means the option value is more affected by changes in volatility.
- **Rho** (ρ) — Measures the sensitivity of the option price to changes in the risk-free interest rate.

2.4 Analysis and Visualization

To study the behavior of these sensitivities:

- We compute and visualize **Delta** and **Gamma** as functions of both the stock price S_0 and volatility σ .
- These surfaces illustrate how hedging sensitivity and curvature evolve across different volatility regimes and underlying prices.

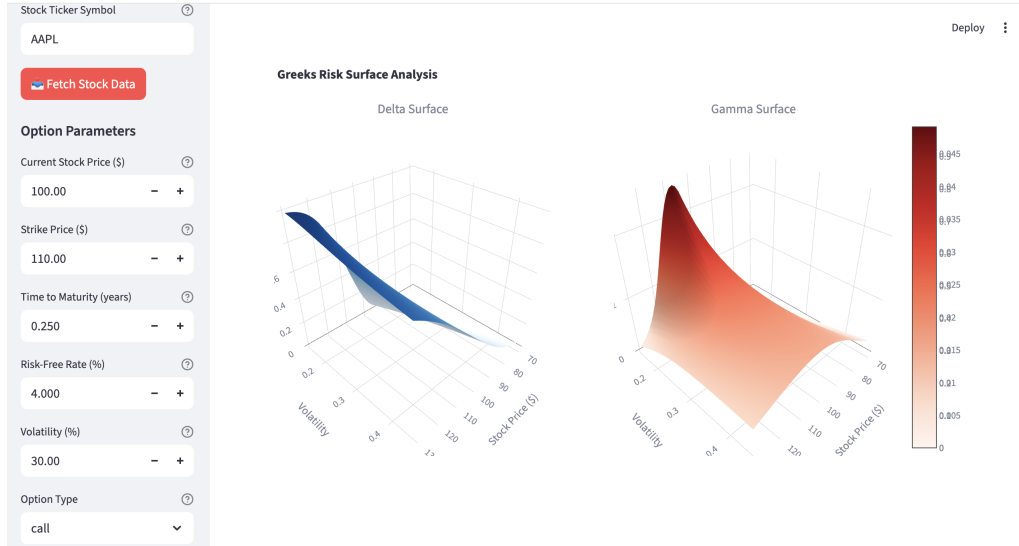


Figure 4: Variation of Δ and Γ with respect to stock price and volatility for a given input

2.5 Key Insights

- Delta increases smoothly from 0 to 1 as the option moves from deep out-of-the-money to deep in-the-money.
- Gamma peaks near-the-money, showing maximum sensitivity of Delta when $S_0 \approx K$.
- At higher volatility, Gamma flattens, reflecting smoother option price behavior across stock prices.

3 Tree Visualization

3.1 Objective

This section presents a visual and analytical understanding of the **Binomial Option Pricing Model**. The model constructs a recombining tree of possible stock prices over discrete time intervals and computes the option price by backward induction. The goal is to visualize the **evolution of stock prices, option values, and hedge ratios (Delta)** within the lattice.

3.2 Model Setup and Assumptions

The binomial model assumes:

- Trading occurs in discrete time steps of equal length.
- The underlying asset price can move up or down by fixed factors in each step.
- Markets are frictionless (no transaction costs, perfect divisibility, continuous compounding).
- No arbitrage opportunities exist.

- The risk-free rate r and volatility σ are constant over time.

3.3 Formulas

For a time horizon T divided into N equal steps of length $\Delta t = \frac{T}{N}$:

1. Up and Down Factors The stock price can move:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$$

so that after i up moves and j down moves ($i + j = N$):

$$S_{i,j} = S_0 u^i d^j$$

2. Risk-Neutral Probability Under no-arbitrage conditions, the equivalent martingale measure gives:

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

where p is the probability of an up move in the risk-neutral world.

3. Option Valuation by Backward Induction At maturity ($t = T$):

$$C_{N,i} = \max(0, S_{N,i} - K)$$

For earlier nodes ($t = t_j$):

$$C_{j,i} = e^{-r\Delta t} [p C_{j+1,i+1} + (1 - p) C_{j+1,i}]$$

4. Hedge Ratio (Delta) The hedge ratio $\Delta_{j,i}$ at each node measures sensitivity of the option to the underlying:

$$\Delta_{j,i} = \frac{C_{j+1,i+1} - C_{j+1,i}}{S_{j+1,i+1} - S_{j+1,i}}$$

3.4 Interpretation

- u and d define how much the stock price can rise or fall per time step.
- p is not a real probability — it ensures no arbitrage and that expected discounted prices follow the risk-free rate.
- The model converges to the Black–Scholes price as $N \rightarrow \infty$.
- The hedge ratio Δ provides the number of shares required to replicate the option at each step.

3.5 Visualization

We generate a lattice (binomial tree) showing:

- Stock price evolution at each node.
- Option price computed backward from maturity.
- Corresponding hedge ratios (Δ) at each node.

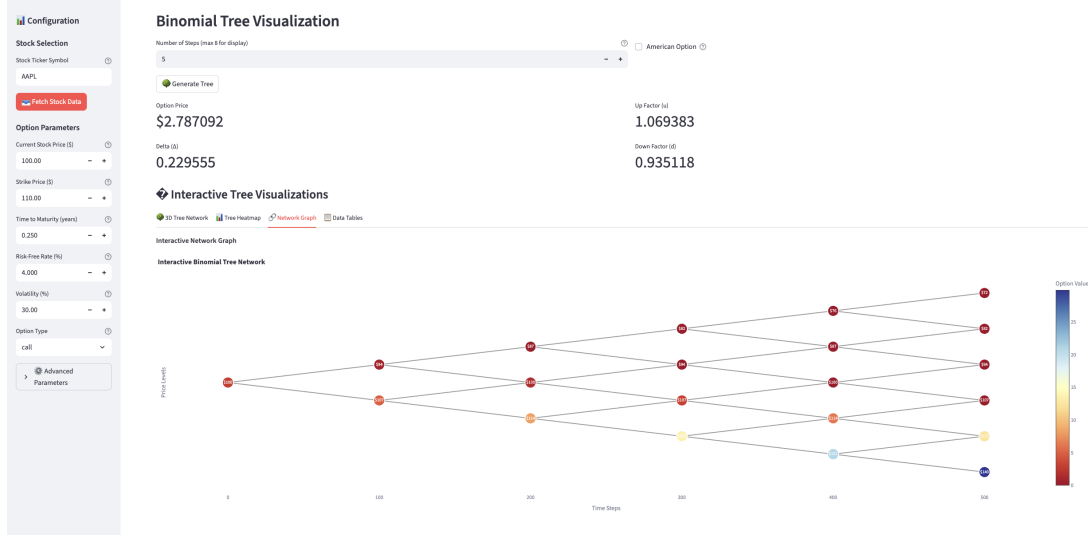


Figure 5: Binomial Tree Visualization showing stock prices and option prices across nodes.

3.6 Key Insights

- The lattice provides an intuitive discrete approximation of stochastic price evolution.
- Option value at each node represents the fair value under risk-neutral probabilities.

4 Option Payoffs

4.1 Objective

This section presents the payoff structures of common options and option combinations. We consider basic positions (long/short calls and puts) and multi-leg strategies (spreads, straddles, butterflies) to illustrate their profit/loss profiles as a function of the underlying asset price at maturity.

4.2 1. Basic Option Positions

Long Call (LC)

$$\Pi_{LC}(S_T) = \max(S_T - K, 0) - C_0$$

Right to buy at strike K . Profit unlimited above $K + C_0$, loss limited to premium C_0 .

Short Call (SC)

$$\Pi_{SC}(S_T) = C_0 - \max(S_T - K, 0)$$

Obligation to sell if exercised. Maximum gain is premium C_0 ; unlimited loss if $S_T \gg K$.

Long Put (LP)

$$\Pi_{LP}(S_T) = \max(K - S_T, 0) - P_0$$

Right to sell at strike K . Profit when $S_T < K - P_0$, loss limited to premium P_0 .

Short Put (SP)

$$\Pi_{SP}(S_T) = P_0 - \max(K - S_T, 0)$$

Obligation to buy if exercised. Maximum gain is premium P_0 ; loss increases as $S_T \rightarrow 0$.

4.3 2. Vertical Spreads

Bull Call Spread (BCS) Buy a call at K_1 , sell a call at $K_2 > K_1$:

$$\Pi_{BCS}(S_T) = \max(S_T - K_1, 0) - \max(S_T - K_2, 0) - (C_1 - C_2)$$

Limits upside and downside; bullish outlook.

Bear Put Spread (BPS) Buy a put at K_2 , sell a put at $K_1 < K_2$:

$$\Pi_{BPS}(S_T) = \max(K_2 - S_T, 0) - \max(K_1 - S_T, 0) - (P_2 - P_1)$$

Profitable when stock falls; loss limited to net premium.

4.4 3. Straddles

Long Straddle (LS) Buy a call and a put at same strike K :

$$\Pi_{LS}(S_T) = \max(S_T - K, 0) + \max(K - S_T, 0) - (C_0 + P_0)$$

Profitable for large movements in either direction; loss limited to total premium.

Short Straddle (SS) Sell a call and a put at strike K :

$$\Pi_{SS}(S_T) = (C_0 + P_0) - \max(S_T - K, 0) - \max(K - S_T, 0)$$

Profitable for stable markets; unlimited risk if large moves occur.

4.5 4. Butterfly Spread

Long Butterfly Spread (Call) Buy calls at K_1 and K_3 , sell two calls at $K_2 = \frac{K_1 + K_3}{2}$:

$$\Pi_{BF}(S_T) = \max(S_T - K_1, 0) - 2\max(S_T - K_2, 0) + \max(S_T - K_3, 0) - \text{net premium}$$

Profitable when S_T near K_2 ; low risk, limited profit.

4.6 5. Visualization

- All strategies are plotted on a single graph: profit/loss ($\Pi(S_T)$) versus terminal stock price S_T .
- Helps compare risk, reward, and break-even points across different option strategies.

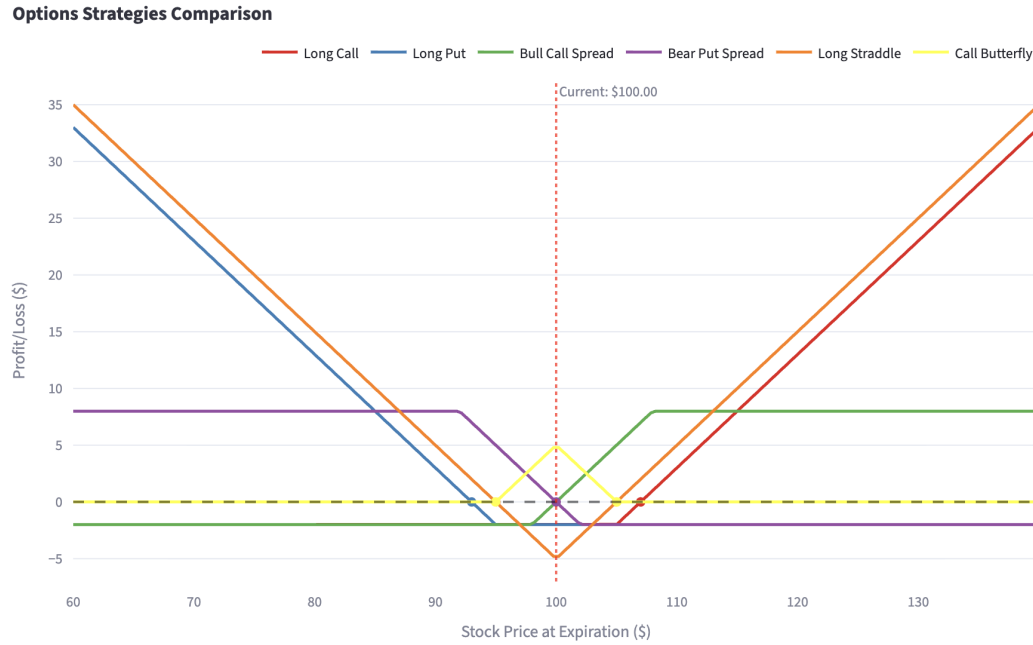


Figure 6: Unified payoff diagram for long/short call, long/short put, bull/bear spreads, straddles, and butterfly spread.

Strategy Comparison Table

	Strategy	Max Profit	Max Loss	Breakeven Points	Net Premium	Risk/Reward
0	Long Call	\$33.00	\$2.00	1	\$2.00	0.06
1	Long Put	\$33.00	\$2.00	1	\$2.00	0.06
2	Bull Call Spread	\$8.00	\$2.00	1	\$2.00	0.25
3	Bear Put Spread	\$8.00	\$2.00	1	\$2.00	0.25
4	Long Straddle	\$35.00	\$4.80	2	\$5.00	0.14
5	Call Butterfly	\$4.80	\$0.00	2	\$0.00	0.00

Figure 7: Strategy Comparison Table for a sample input

5 Convergence Analysis

5.1 Objective

This section studies how the discrete numerical approximation given by the Cox–Ross–Rubinstein (CRR) binomial model converges to the continuous-time Black–Scholes–Merton

(BSM) price as the number of binomial steps N increases. We present both the *price convergence plot* and the *error plot* (absolute and relative error) for a representative input set up to $N = 1000$.

5.2 Setup and notation

Let

C_{BSM} = Black–Scholes–Merton call price, $C_{CRR}(N)$ = CRR call price computed with N steps.

We study the sequence $\{C_{CRR}(N)\}_{N \geq 1}$ and its deviation from C_{BSM} as $N \rightarrow \infty$.

5.3 Error metrics

Define the following error measures used in the plots:

$$\text{Absolute error:} \quad \Delta_{\text{abs}}(N) := |C_{CRR}(N) - C_{BSM}|.$$

$$\text{Relative error:} \quad \Delta_{\text{rel}}(N) := \frac{|C_{CRR}(N) - C_{BSM}|}{C_{BSM}}.$$

For reporting numerical stability, also consider the signed error

$$\Delta_{\text{signed}}(N) := C_{CRR}(N) - C_{BSM},$$

which shows whether the discrete model over- or under-estimates the benchmark.

5.4 Qualitative behaviour and explanation

- **Convergence intuition:** The CRR model is a discrete recombining approximation of geometric Brownian motion. As the step size $\Delta t = T/N \rightarrow 0$ (i.e., $N \rightarrow \infty$) the discrete distribution of the log-price approaches the continuous log-normal distribution assumed by BSM, and the CRR price approaches C_{BSM} .
- **Monotonicity (practical observation):** Depending on parameter choices and implementation, $C_{CRR}(N)$ may oscillate slightly for small N and then settle toward the BSM value; in many practical parameter regimes the sequence displays smooth approach with diminishing error.
- **Why examine up to $N = 1000$:** This range shows both the small- N behaviour (where discretization error is visible) and the large- N region where the approximation is practically indistinguishable from BSM for typical option parameters.
- **Computational considerations:** Larger N increases computational cost and memory usage (though for recombining trees one can use linear memory via backward induction). The convergence plots let you assess the trade-off between accuracy and computational effort.

5.5 Plot descriptions (for a sample input set)

For a chosen example parameter set (e.g., $S_0 = 100, K = 110, r = 0.04, \sigma = 0.30, T = 0.25$), The CRR Price Convergence and CRR Error Convergence plots looks as below

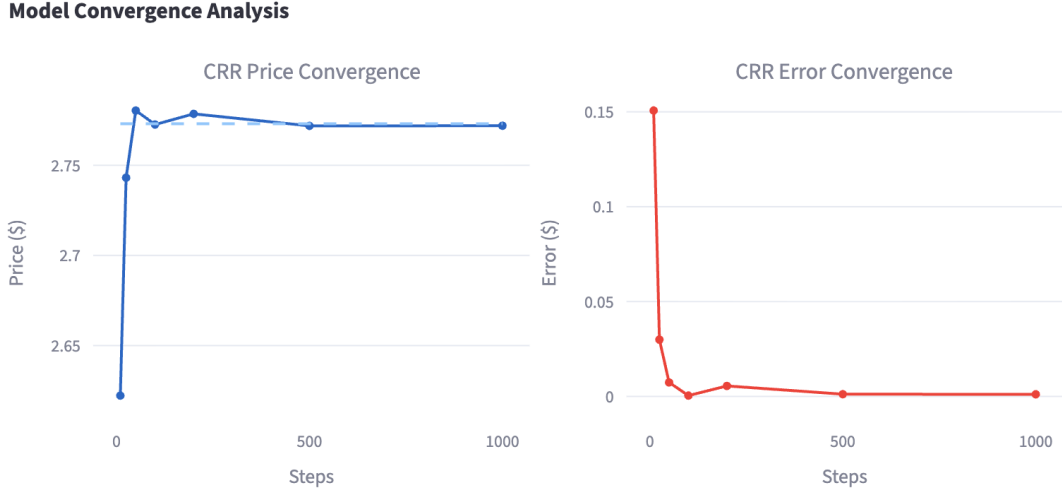


Figure 8: Convergence of CRR binomial prices $C_{CRR}(N)$ toward the BSM benchmark C_{BSM} and CRR Error as N increases (example input set).

5.6 Monte Carlo vs Binomial remarks

Although this section focuses on CRR convergence, you may juxtapose the binomial error behaviour with Monte Carlo sampling error for completeness:

- Monte Carlo error decays in expectation like $O(M^{-1/2})$ where M is the number of simulated paths. Use confidence intervals to show statistical uncertainty.
- Binomial discretization error is deterministic given N (no sampling noise) and can be reduced by increasing N .

6 Trading Strategies

6.1 Objective

This section describes and analyzes various option-based trading strategies implemented in the project. Each strategy combines option and stock positions to achieve specific risk–return profiles, and their performance is compared to a simple buy-and-hold benchmark.

6.2 1. Covered Call Strategy

A *covered call* involves holding a long position in the underlying stock while simultaneously writing (selling) a call option on the same asset.

$$\text{Portfolio} = +1 \text{ Stock} - 1 \text{ Call Option.}$$

Payoff at maturity:

$$\Pi_{CC}(S_T) = \begin{cases} (S_T - K) + C_0, & S_T < K, \\ (K - K) + C_0 = C_0, & S_T \geq K, \end{cases}$$

where C_0 is the call premium received.

Interpretation:

- Generates extra income through the option premium.
- Limits upside potential beyond strike K .
- Suitable for mildly bullish or neutral outlooks.

6.3 2. Long Straddle Strategy

A *long straddle* profits from large price movements (either upward or downward) by combining a long call and a long put with the same strike and maturity.

$$\text{Portfolio} = +1 \text{ Call Option} + 1 \text{ Put Option.}$$

Payoff at maturity:

$$\Pi_{\text{Straddle}}(S_T) = \max(S_T - K, 0) + \max(K - S_T, 0) - (C_0 + P_0),$$

where C_0 and P_0 are the premiums paid for the call and put, respectively.

Interpretation:

- Profitable for high volatility scenarios.
- Loss limited to total premium paid.
- Break-even points: $S_T = K \pm (C_0 + P_0)$.

6.4 3. Delta-Neutral Strategy

A *delta-neutral portfolio* hedges directional exposure by offsetting the option's delta with an opposite position in the underlying stock.

$$\Delta_P = \Delta_{\text{option}} + n\Delta_{\text{stock}} = 0.$$

For a single call option,

$$n = -\frac{\Delta_{\text{option}}}{\Delta_{\text{stock}}} = -\Delta_{\text{option}},$$

since $\Delta_{\text{stock}} = 1$.

Interpretation:

- Neutralizes small price changes in the underlying.
- Requires dynamic rebalancing as Δ changes over time (gamma risk).
- Used for short-term hedging or market-making.

6.5 4. Buy-and-Hold Benchmark

This benchmark strategy involves buying the underlying asset and holding it until maturity, with no option positions or hedging adjustments.

$$\Pi_{\text{BH}}(S_T) = S_T - S_0.$$

Interpretation:

- Represents the pure market exposure.
- Serves as a baseline for comparing risk-adjusted performance of option-based strategies.

7 Hedging Simulation

7.1 Objective

This section describes the simulation framework for implementing **delta-neutral** and **gamma-neutral** hedging strategies. The goal is to maintain a portfolio that is locally insensitive to small movements in the underlying asset and analyze the resulting profit and loss (P&L) distribution.

7.2 Hedging Concepts

Delta Hedging

- A delta-neutral portfolio is constructed to offset the option's sensitivity to the underlying price:

$$\Delta_P = \Delta_{\text{option}} + n\Delta_{\text{stock}} = 0 \quad \implies \quad n = -\Delta_{\text{option}}$$

where n is the number of shares held (or shorted) in the underlying.

- The portfolio is rebalanced at discrete time intervals, Δt , known as the *hedging frequency*.
- Objective: eliminate first-order exposure to price movements.

Gamma Hedging (optional)

- Gamma hedging extends delta-neutral hedging to reduce curvature risk:

$$\Gamma_P = \Gamma_{\text{option}} + n_\Gamma \Gamma_{\text{hedging instrument}} = 0$$

- Requires additional instruments (options) to offset the second-order sensitivity.
- Gamma hedging reduces the risk of large deviations in the underlying asset, complementing delta hedging.

7.3 Simulation Framework

- Simulate M paths of the underlying asset price S_t under the risk-neutral dynamics:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

using discrete time steps.

- At each time step:
 1. Compute current option delta (and gamma if applicable).
 2. Adjust the portfolio by buying/selling underlying stock (and/or options) to maintain neutrality.
 3. Record the portfolio value after rebalancing.
- Repeat until maturity T , then compute the final portfolio P&L:

$$\text{P\&L} = V_T - V_0$$

7.4 Performance Metrics

After simulating multiple paths, compute statistical measures:

- **Mean P&L:**

$$\mu_{\text{P\&L}} = \frac{1}{M} \sum_{i=1}^M \text{P\&L}_i$$

- **Standard deviation of P&L:**

$$\sigma_{\text{P\&L}} = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (\text{P\&L}_i - \mu_{\text{P\&L}})^2}$$

- **Maximum profit:** $\max_i \text{P\&L}_i$
- **Maximum loss:** $\min_i \text{P\&L}_i$

7.5 Interpretation

- Delta and gamma hedging aim to minimize exposure to market movements and volatility risk.
- Performance metrics quantify both the effectiveness and the risk of the hedging strategy.
- Hedging frequency directly impacts P&L variability: higher frequency reduces risk but increases transaction costs in practice.
- Neutral strategies ideally produce mean P&L close to zero with low standard deviation, reflecting effective risk management.